Exploring the readiness for deployment of AI-driven Military Systems in Africa: Evidence from Nigeria, Kenya, and Burkina Faso

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ABSTRACT

Advancements in artificial intelligence (AI) are significantly redefining warfare, with researchers suggesting the possibility of a global arms race as advanced militaries are increasingly integrating AI into their weaponry systems and increasing investments in AI development. Although African militaries are not directly involved in developing AI-driven systems, research over the past five years reveals an increase in the importation and deployment of such systems by African militaries, particularly in conflict zones. However, the absence of robust, context-specific AI governance frameworks that articulate the responsible adoption of AI-driven systems in African militaries is a major concern, as this results in a high likelihood of unpredictable responses in combat, leading to civilian fatalities and serious human rights violations. Therefore,through an exploratory qualitative case study research design, this study analysed the impact of deploying AI-driven systems by African militaries in conflict zones in the absence of robust AI governance frameworks. Nigeria, Burkina Faso and Kenya were the case studies and model variables investigated were transparency, safety, accountability, and security. The research revealed the policy positions of the case studies on the deployment of AI-driven systems with reference to the model variables explored.The study proposed several measures, including the creation of a regionally harmonised AI governance framework that prioritises transparency, security, accountability, and safety in the African military domain by the African Union. It also recommends developing repositories for AI-driven systems deployed in the military domain in Africa to ensure accountability.

Keywords: Artificial Intelligence, AI governance, military domain, Africa, policy, safety, and accountability.

1. Introduction

Since World War II, there has not been such a significant increase in military budgets by advanced militaries globally as has been witnessed over the past five years. Research identifies artificial intelligence (AI) as the main driver of these developments. Whereby advanced militaries are significantly increasing their budgets towards AI research and development to enhance production and integration of AI in their military systems, with an aim of improving their military capabilities and maintaining their competitive advantage over their adversaries [1], [2]. Although this is becoming a global trend, Africa finds itself either as a consumer or a testing ground for AI-driven military systems resulting from the research and development activities of advanced militaries. Unfortunately, this is occurring in an environment with underdeveloped AI governance frameworks, inadequate technological infrastructure and expertise [3], [4], [5], [6].

The absence of both international and regional AI governance frameworks erodes the benefits that AI-driven systems could offer African militaries. These benefits include improved logistics, strategic decision-making, and data processing among others [7], [8]. Instead, presents high risks, such as unpredictable responses to conflicts, human rights violations and opaque decision-making processes.

Considering Africa's complex security environment that is characterised by conflicts and insurgencies, successful deployment of AI-driven military systems would require robust AI governance frameworks that address AI ethics dimensions such as safety, security, accountability, and transparency. Given that countries such as Nigeria, Burkina Faso in the Sahel, and Kenya in East Africa, are currently deploying AI-driven systems in their conflict zones without robust regional or international AI governance frameworks makes them suitable case studies for examining the impact of uncontrolled adoption of AI-driven systems in African conflict zones [4], [9]. Therefore, this study explored the impact of deploying AI-driven systems by African militaries operating in conflict zones.

1.1 Problem statement

Due to significant advances in AI developments advanced militaries such as those of the US, China, Russia, Israel, and the UK are increasingly investing in the research and development of

AI-driven systems. In contrast, smaller powers, including African militaries, are compelled to rely on the importation of these AI-driven systems to enhance their military capabilities. In Africa, the procurement and deployment of these systems is highly complex, as insurgent groups also seek to improve their military capabilities through the illegal importation of AI-driven systems. Another concern is the possibility that African conflict zones may be used as testing grounds by major powers for their newly developed AI-driven systems. Of particular concern is the adoption of AI-driven systems by both government militaries and insurgent groups in environments characterised by limited AI governance frameworks for the military domain. Risks in such environments include unpredictable or disproportionate responses that may undermine efforts to resolve conflict, excessive human rights violations with limited civilian oversight, as transparency is lacking and accountability mechanisms are absent. Therefore, this study, through a qualitative case study research design, investigates the degree of adherence to the AI ethics dimensions of transparency, safety, security, and accountability by the case study countries: Nigeria, Burkina Faso in the Sahel, and Kenya in East Africa.

1.2 Research Questions

The following research questions guided the study:

RQ1: Which AI-driven systems and lethal autonomous weapons are deployed in Nigeria, Burkina Faso, and Kenya, and how do they incorporate ethical, safety, and security safeguards?

RQ2: Which actors or organisations are responsible for these AI systems, and what accountability and transparency mechanisms govern their deployment?

RQ3: In which operational contexts are these AI systems used (eg, logistics and decision-making), and how are ethical and safety standards maintained?

RQ4: To what extent have these AI systems contributed to or mitigated human rights violations, and how are such violations monitored and addressed through transparent and accountable mechanisms?

This article is organised as follows. Section 2 presents a literature review on this topic, followed by Section 3, which outlines the research method. The results and discussions are presented in Section 4, Section 5 contains the recommendations, and Section 6 concludes the paper.

2. Literature Review

The advancement of AI integration into defence technologies is revolutionising the landscape of modern warfare. The researchers herein; examined the theoretical framework underlying the development of these military capabilities along the spectrum of autonomy. This spectrum categorizes the continuum of human control over perception, decision-making, and action in military systems; scoping the purely human-controlled tools, to independent AI-controlled platforms.

The five recognised phases of this spectrum, namely: manual phase, operator assistance phase, task-based autonomy, conditional autonomy, and highly autonomous systems; demonstrate the historical and technological shift from completely human involvement to maximum machine independence [10]. In the manual phase deploying the mechanical systems of the First and Second World Wars ; complete human control was exercised. Albeit at a high operational risk [11],[12].

The introduction of technologies, such as autopilot and early fire control systems; marked the launch of the operator assistance phase. Herein; automation enhanced support and the human workload was reduced [13],[14],[15]. Humans did however maintain primary decision-making capacity. Subsequent advances led to the task autonomy phase; which was characterised by "set-and-forget" systems, which operate independently and execute predetermined missions[16],[17]. Recent advances in technology have instilled conditional autonomy[18],[19]. This phase has led to the development of automated military systems; which consist of advanced Unmanned Aerial Vehicles (UAVs) like the MQ-9 Reaper; as well as active protection systems capable of making decisions within predefined limits, all the while generally remaining under human supervision. As of now, the emergence of highly autonomous systems, such as Lethal Autonomous Weapon Systems (LAWS), and experimental autonomous fighter jets; represents a significant paradigm shift. Therein; systems have the ability to make independent, complex decisions within human-defined objectives; and shift human control from direct operation to strategic oversight [20],[21].

Understanding this evolution of the autonomy spectrum enables just insight into the progression of military systems towards LAWS; particularly, their operational impact should the militaries of Nigeria, Kenya, and Burkina Faso deploy them in their respective conflict zones. The progression of the autonomy spectrum also marks a baseline for current military technological capabilities. It further provides the necessary context to address ethical and compliance challenges that are associated with increasing delegation of critical military human capacity to machines in the African context.

2.1 Global debates on AI in military applications.

Lethal autonomous weapon systems (LAWS) drive deliberations on AI-driven military systems in the global community. Countries are divided with regard to the resolve to constrain these systems outright, or to enforce a framework to maximize on their benefits while minimizing risk exposure[22],[23].This ongoing stalemate highlights the difficulty balancing between technological innovation, military edge, and humanitarian customs. Researchers have amplified the ethical dilemma of delegating life-and-death decisions to machination; stating a significant distention with the existing frameworks of international humanitarian law which raise questions regarding the erosion of human judgment in warfare [24],[25].

As the handling of LAWS has become highly topical; numerous forums on the responsible use of AI in the military are regularly convened. Therein, emphasising the global importance of the matter . In the 2022 expert workshop on the responsible use of AI in military systems, organised on behalf of the Dutch Ministry of Defence[26],[27],[28], a phase of intensive engagement with the topic is instigated. This platform brought together experts from academia, the public and private sector, and research organisations for an interactive session. The deliberations dwelt on the development, deployment, and use of AI in the military sector in an ethically justifiable manner, accounting for the rapid progress of the technology.

In 2023, the first international summit on Responsible AI in Military Systems (REAIM) was held in The Hague, Netherlands, jointly organised by the Dutch Ministry of Foreign Affairs and the Ministry of Defence[26]. This event marked the start of a global initiative to promote the responsible use of AI in the military domain. In 2024, the second REAIM Summit was co-hosted by the Republic of Korea, Singapore, the Netherlands, Kenya, and the United Kingdom on 9 and 10 September in Seoul[29]. Regardless of significant and minor differences among military leaders concerning the introduction of responsible AI; the second REAIM Summit resulted in a Blueprint for Action. This document builds on the call to act from the previous year's summit in The Hague. The aforementioned Blueprint expands on protocols for logistics, intelligence, operations, and acceptable logic for the introduction of AI in the armed forces [30]. As of the date of adoption, 60 countries, including our subject, Kenya, have endorsed the Blueprint [31].

The Blueprint for Action agreement was adopted in a global environment where conflicts involving AI-driven lethal autonomous weapons were capturing global attention; particularly in the disputes between Ukraine and Russia, Palestine and Israel, and in Libya[32].Deployed systems have consisted of Saker Scout, Gospel, and Kargu-II. Countries including China, Israel, Russia, South Korea, Turkey, the United Kingdom, and the United States; are reported to be investing in the development of AI-driven autonomous weapons [33],[34],[35],[36],[37]. However, the use of AI-driven systems in conflicts go beyond those mentioned above. Despite growing global attention; a significant gap in scholarship, access to public information regarding the specific AI-driven systems used by armed forces, insurgents, and rebels in various African conflicts, is prevalent. While details are limited; the evidence of such technologies being used in regions such as the Sahel, particularly in Nigeria and Burkina Faso, as well as in East Africa, including Kenya, exists for discovery[3].

2.2 Principles of Military Ethics

The principles of military ethics provide a framework which can regulate behaviour in armed conflict; with emphasis on the protection of civilians and accountability [38]. However, when AI-driven military systems are deployed; unique challenges regarding the principles of distinction, proportionality, and responsibility, as autonomous technologies complicate contextual judgement and accountability; emerge. The scarcity of AI governance frameworks that deliberately articulate conduct in the military, further exacerbates these challenges in Africa, making it difficult for African militaries to comply with international humanitarian law in countries such as Nigeria, Kenya, and Burkina Faso [39].This section addresses these challenges and highlights the need for an AI governance framework adapted to local realities in the military domain.

The relationship between military ethics and the integration of AI-driven systems in the military is established on the role of ethical principles as the normative foundation for evaluating the deployment of AI-driven systems. Military ethics form the moral parameters within which these technologies should operate; with emphasis on the principles of distinction, proportionality, and accountability. In many respects, however, AI-driven systems cannot enforce these principles. In some instances; AI-driven systems cannot satisfy the requirement of distinction due to limitations of contextual perception and moral reasoning [40]. Similarly; AI-driven systems cannot adequately execute the principle of proportionality, due to a lack of capacity for nuanced moral judgement [41]. Accountability, when ethical or legal violations occur, is difficult to ascertain as a result of the distribution of responsibility among system developers, military commanders, and the autonomous systems themselves [42].

In Africa, military ethical principles application in situations where AI-driven systems are deployed in conflict zones, remains unfeasible due to weak and limited oversight mechanisms, fragmented AI governance structures, and a lack of technical expertise. These conditions breed ethical risks; such as civilian casualties, violation of international humanitarian law (IHL), and increased exposure to cyber threats [43],[44]. In contrast to the Global North, with robust regulatory systems; African states lack the institutional capacity to ensure compliance with military ethics when using AI-driven systems in the military; particularly in conflict zones. This escalates concern that the use of AI-driven systems in the military may instigate further insecurity in the continent rather than promote its stability.

Transparency encourages open practices in the procurement and deployment of AI systems within military doctrine. Safety that requires rigorous testing and a commitment to complete human oversight throughout its use; is critical. Accountability is important as it clearly defines and assigns responsibility for all outcomes; especially in the event of malfunctions or violations. Security, which needs a balanced execution to protecting the state while safeguarding human safety and the needs of the civilian population, would ensure confidence in AI-driven systems deployed in conflict zones[45].

The research case studies echo the urgency of these principles. In Nigeria; drone operations against Boko Haram and ISWAP are expanding without transparent procurement protocols; and have limited civilian oversight[46],[47]. Consequently; accountability is scarce. Kenya has instituted a strict civilian data protection regime [48],[49];.However, the regime lacks safeguards on the application of AI-driven systems in the military; thus raising ethical concerns in the context of counter-terrorism surveillance. Burkina Faso is reported to have deployed Bayraktar TB2 drones in the Sahel. The lack of a clear targeting system and weak accountability mechanisms has reportedly resulted in civilian casualties [50].These cases demonstrate how AI governance frameworks for deployment of AI-driven systems in the African militaries require more than simply the adoption of global norms. There is an urgency for a context-specific AI governance framework that speaks to local realities; with stronger civil-military oversight and integration of military ethics into institutional practices; in order to prevent escalation, abuse, and harm to vulnerable populations.

2.3 AI Governance and Military Ethics

Since AI governance principles and military ethics are intertwined [30][51], to effectively manage the risks arising from AI-driven military systems, the African Union should design AI governance systems that embody both AI ethics dimensions and military ethics. Such a strategy ensures compliance with international law and the protection of civilians.

Furthermore,the use of AI-driven systems in the military also includes country-specific factors; ranging from national laws, to procurement procedures, and operational context. The strength of a country's legal framework for data and procurement, as well as its institutional and doctrinal guidelines for AI-driven systems in the military; will determine the degree to which these systems are deployed responsibly, and the adequacy of the protection of the civilian population. According to research conducted by SIPRI and UNIDIR, a lack of transparency, safety, accountability, and security are exacerbated in countries with lower governance capacities; highlighting the importance of conducting country-level analyses[52],[53].

2.4 Gaps in transparency, safety, accountability, and security in African militaries.

This subsection synthesises literature on four interrelated military AI governance deficits: transparency, operational safety, legal and institutional accountability, and national security, which recur in analyses of military uses of AI-driven systems (notably unmanned aerial systems, automated sensing, and decision-support tools) in Africa. The review highlights structural causes and specific harms drawing on official government documents(legislation & white paper), international policy reports, NGO investigations, and regional AI governance frameworks assessments.

Transparency gaps

A persistent finding in the literature is that military procurement, capability disclosure, and operational doctrine in many African states are opaque by design or necessity [54]. Additionally, military budgets, procurement contracts for high-impact systems (e.g., MALE drones), and rules of engagement are often not publicly available or are classified, which constrains independent risk assessment and prevents timely public accountability [55]. Analyses of the governance of AI-driven systems deployed in the military, as seen in the case studies, reveal weak oversight of procurement and secrecy surrounding defence expenditure and capabilities, which complicates civil society and parliamentary scrutiny of emerging AI-driven systems [53].

The lack of transparency has led external monitoring groups to rely on open-source evidence and post-incident reports to track harms caused by the deployment of AI-driven systems in these conflict zones under study. Furthermore,the rapid and sometimes poorly documented acquisition of MALE and armed UAS by states, sometimes from non-traditional exporters, has outpaced public doctrine and transparency mechanisms, leaving civil society and international bodies with limited ability to evaluate safety controls [56], [57].

Safety and operational control deficiencies

Literature on armed drones and AI-driven targeting emphasises that safety depends not only on technical reliability but also on doctrine, human-in-the-loop safeguards, testing regimes, and red-teaming prior to deployment[58],[59]. Where programmes lack publicised testing standards, independent validation, or clear policies on human oversight, the risk of misidentification, erroneous targeting, and accidental escalation increases, especially in complex counter-insurgency environments common in the Sahel and parts of West Africa [60],[61]. The technical fragility of some machine learning systems in unstructured, noisy operational environments further heightens these risks where institutional safety practices are weak or absent [62].

Drone strikes causing civilian casualties in Burkina Faso and neighbouring countries have been reported by independent investigators and NGOs. Issues highlighted include a lack of expertise in deploying AI-driven systems, resulting in civilian fatalities, and the absence of reporting mechanisms on post-strike incidents in the region regarding the harms caused by military operations [63], [3].

Inadequate Accountability

Tracing responsibility for harms arising from AI-driven systems in African conflict zones is currently not feasible, as existing chains of command and investigatory mechanisms lack provisions to address issues resulting from machine-assisted decisions [64], [65]. Reports by human rights organisations on abuses in African conflict zones reveal a lack of accountability for unlawful killings and excessive ill-treatment of civilians by security forces, where tracking the perpetrators has not been possible. This also applies to harms caused by AI-driven systems deployed in these conflict zones [4], [66]. Complex political dynamics, such as coups, exacerbate the difficulties in tracing perpetrators and bringing them to account. Recent reports from the Sahel highlight how governance breakdowns and politicised security institutions can hinder accountability processes [67], [68].

Security and technical expertise

Effective oversight of AI-driven military systems requires robust socio-technical systems [65][69]. However, this is highly problematic in Africa, as reports have emerged of oversight bodies being under-resourced, if they exist at all, in several member states, coupled with limited expertise in the field of AI. This situation makes it difficult to enforce standards, demand remedial action, or participate effectively in international norm-building [70][71]. Conversely, dual-use supply chains and the international spread of drone technology enable states to acquire advanced capabilities more quickly than domestic oversight institutions can absorb the knowledge required for effective regulation and auditing. Researchers on proliferation dynamics recommend parallel investments in institutional capacity, standards, and regional cooperation to address this gap [72][58].

Scholars and policy analysts have identified reasons that link the four gaps above: thus (1) procurement and operational secrecy; (2) insufficient doctrine on meaningful human control and operational limits for AI-driven systems; (3) weak resourcing and expertise within oversight institutions; and (4) rapid external acquisition of capability without corresponding domestic safeguards. Together, these factors increase the risk that AI-driven systems deployed in the military will cause civilian harm, violate IHL, and erode public trust in security institutions, [73],[35],[74].

2.5 Research and Knowledge Gaps

As highlighted in the GPAI report (2024), several governments at the supranational, national, and subnational levels, along with civil society organisations and universities, have established repositories that provide information on automated decision-making systems and AI-driven systems used by public institutions. Some of these repositories also include systems used by the military and law enforcement agencies. However, no such repositories have been identified on the African continent [75]. There is thus a notable gap in understanding how public agencies in African countries use AI-driven systems. Furthermore, there is a more specific gap regarding AI-driven systems used for military purposes. The use of AI-driven systems in intra- or inter-state conflicts is not limited to the armed forces; armed rebels and insurgent groups also use such technologies.

A third knowledge gap identified in this study concerns the current academic literature on the use of AI-driven systems for military purposes in Africa. Existing publications generally focus on the following main areas: security[76], geopolitical competition[77], applications and implications of deploying AI-driven systems in the military domain[78], and lessons from other regions to inform future deployment of AI-driven systems on the continent[79]. These contributions inform policy debates but do not address the consequences of deploying AI-driven systems in conflict zones where there are no AI governance frameworks or guidelines for their deployment.

Therefore, this study seeks to address these gaps in an area that remains largely unexplored on the continent.

3. Research Method

The study employed qualitative exploratory case study research design. The research questions outlined in section 1.2 guided the data collection process which was conducted through a comprehensive literature review along with analysis of official government documents and international reports. Google scholar, Scopus and Sciencedirect search engines were utilised and keywords such as “artificial intelligence”, “military” “conflict” ,”Nigeria”, “Burkina Faso” , “AI-driven systems” provided a strategy to locate publicly available reports.A systematic search for reports from international organisations and academic papers that specifically reports on deployment of AI-driven systems by the militaries of the countries under study was conducted. This involved reviewing documents from institutions such as the United Nations Institute for Disarmament Research, Human Rights Watch, Drone Wars, and peer-reviewed academic publications. These sources provided valuable context and, in some cases, concrete examples of AI applications in African conflict zones.

The study then progressed from comparative qualitative analysis to quantitative causal inference using a Regression Discontinuity (RD) design, with R as the programming language. As a classic RD model requires a large sample of observations clustered around a sharp policy cutoff, applying it directly to only three countries is infeasible. Therefore, the following models are presented as hypothetical Sharp Regression Discontinuity blueprints. These models use the defined nominal variables as outcome metrics Y and posit a continuous "running variable" X as a proxy for policy capacity or technical maturity, which, upon crossing a threshold c, triggers a theoretical binary policy change (the "treatment").

General Regression Discontinuity Framework was deployed where all models are based on a local linear regression approach, focusing only on observations near the cutoff point c. This structure isolates the causal effect (τ) of the binary policy change D_i:
Where:

Y_i : The Outcome Variable (measured policy/operational state).

X_i : The Running Variable (a continuous metric of policy maturity or technical capacity).

c: The Policy Cutoff (the threshold for the policy change D_i).

D_i : The Treatment Indicator (1 if $X_i \geq c$, 0 otherwise).

τ : The Local Average Treatment Effect (LATE), representing the causal discontinuity.

β_1, β_2 : Coefficients allowing the relationship between X and Y to vary on either side of the cutoff.

1. Model for Research Question 1 (Systems, Deployment, and Safeguards)

RQ1: Which AI-driven systems and lethal autonomous weapons are deployed, and how do they incorporate ethical, safety, and security safeguards?

Table 1: Model description for RQ1

Component	Variable	Description
Outcome (Y_1)	Q1_Military_Standard_Adherence (Ordinal 1-3)	The estimated degree of adherence to IHL/Ethical standards in kinetic operations (1=Low, 2=Moderate).
Running Variable (X_1)	National AI Investment Density (e.g., % of GDP allocated to AI R&D/Acquisition)	A continuous metric reflecting the state's prioritization and financial commitment to advanced systems.
Treatment (D_1)	Mandatory AI-Specific Rules of Engagement (RoE)	A policy that only comes into effect when the X_1 investment reaches a high threshold c.

Functions Applied For Model RQ1

The causal effect τ is the difference between the conditional expectation of the outcome just above and just below the threshold c:

Conditional Expectation Function (E): Since the outcome Y_1 is ordinal (1, 2, or 3), an Ordered Probit or Logit function is often used to model the probability of achieving a higher standard score (e.g., Y_1 =3) as a continuous function of X_1. In the RD context, we locally approximate this function using linear regression, relying on the central limit theorem and the local nature of the data to treat the discrete outcome as locally continuous for estimation purposes.

Kernel Weighting Function (κ): The linear model is weighted using a kernel function (e.g., Triangular or Uniform) to assign higher probability to observations closest to the cutoff c. This non-parametric approach minimizes bias from data points further away, mathematically focusing the analysis on the discontinuity at $X_1 = c$.

2. Model for Research Question 2 (Actors, Accountability, and Transparency)

RQ2: Which actors are responsible, and what accountability and transparency mechanisms govern their deployment?

Table 2: Model description for RQ2

Component	**Variable**	**Description**
Outcome Y_2	Q2_Civilian_Oversight_Strength (Ordinal 1-2)	The perceived strength of civilian bodies (e.g., ODPC, IPOA, CIL) to scrutinize security sector AI use (1=Weak, 2=Moderate).
Running Variable X_2	Defense Procurement Secrecy Index (Continuous 0-100)	An inverse metric quantifying the transparency in high-value military technology acquisition (100 = full transparency).

Component	Variable	Description
Treatment D_2	Establishment of an Independent Procurement Scrutiny Body	A policy where crossing the transparency threshold (c) mandates the creation of a powerful independent body with audit access to classified contracts.

Functions Applied For Model RQ2:

Discontinuity in Expectation: The parameter τ estimates the instantaneous, local change in the expected civilian oversight strength ϵ precisely at the policy point c. RD assumes that immediately above and below c, the only difference between observations is the implementation of D_2.

Continuity of the Running Variable Density: A critical prerequisite for valid RD is that the probability density function (PDF) of the running variable $f(X_2)$ must be continuous at c. If the density of X_2 were discontinuous at c, it would indicate that political actors manipulated the procurement transparency index X_2 to sort countries just below the threshold, invalidating the claim that assignment to D_2=1 is quasi-random near c.

3. Model for Research Question 3 (Operational Context and Standards Maintenance)

RQ3: In which operational contexts are these AI systems used, and how are ethical and safety standards maintained?

Table 3. Model description for RQ3

Component	Variable	Description

Outcome (Y_3)	Q4_HR_Impact_Civilian_Fatal (Binary 0/1)	Whether the deployment of kinetic AI systems has resulted in documented mass civilian fatalities (1=Yes, 0=No).
Running Variable (X_3)	System Autonomy Score (Continuous 0-100)	A technical score quantifying the degree of system autonomy in the targeting loop (e.g., from Human-in-the-Loop to Human-on-the-Loop).
Treatment (D_3)	Shift to "Human-on-the-Loop" Operational Doctrine	A military doctrine change triggered at c, where human intervention becomes oversight rather than continuous control.

Probability Calculus Functions Applied

1. **Binomial Outcome (Probability Model):** Since Y_3 is a binary outcome (mass fatality event or not), the appropriate conditional expectation is a **Probability Mass Function (PMF)**, meaning we are modeling the probability P(Y_3 = 1 | X_3). E = P(Y_3 = 1 | X_3) In a local linear RD model, \tau estimates the jump in the *probability* of a mass casualty event precisely at the point where the shift to high autonomy (c) is implemented.

2. **Modeling the Probability Discontinuity:** \tau = \lim_{ X_3 \downarrow c} P(Y_3 = 1 | X_3 - \lim_{ X_3 \uparrow c} P(Y_3 = 1 | X_3) This measures the magnitude of the safety risk increase caused *only* by the autonomy policy shift, controlling for general increases in conflict intensity (X_3). The model assumes the increase in autonomy (X_3) is a continuous progression, but the doctrine shift (D_3) is a sharp, binary event at the threshold c.

4. Model for Research Question 4 (HR Violations, Monitoring, and Redress)

RQ4: To what extent have these AI systems contributed to or mitigated human rights violations, and how are such violations monitored and addressed through transparent and accountable mechanisms?

Table 4. Model description for RQ4

Component	Variable	Description
Outcome (Y_4)	Q4_HR_Redress_Judicial_Clarity (Binary 0/1)	Existence of clear legal or judicial pathways for victims to challenge AI-related algorithmic harm (0=No clarity, 1=Clear pathway/Policy Exists).
Running Variable (X_4)	Judiciary Digital Maturity Index (Continuous 0-100)	A continuous score reflecting the court system's technological readiness and digital competence.

Treatment (D_4)	Mandatory Admissibility Rules for AI-Generated Evidence	A formal policy (like a Judiciary AI Policy Framework) is enacted when the courts reach a certain digital maturity level (c).

Probability Calculus Functions Applied

1. **Causality of Policy Adoption:** RD is particularly suited here because it estimates the causal effect of *adopting the judicial policy* (D_4) on the clarity of redress (Y_4). It controls for the baseline trend of digitalization (X_4) and assumes that countries just below and just above the 90% maturity cutoff are otherwise similar in their need for or ability to provide redress.

2. **Sharp Discontinuity in Policy:** The sharp nature of the cutoff means that crossing c ensures P(D_4=1 | X_4) = 1, and not crossing c ensures P(D_4=1 | X_4) = 0. This deterministic nature of the treatment assignment near c is the foundation for the causal interpretation of \tau in the RD design.

3. **Local Non-Parametric Estimation:** To avoid imposing potentially incorrect global assumptions on the functional relationship between digitalization and judicial clarity, the local linear regression employs non-parametric methods. This involves weighting the small sample of data points near the threshold to accurately capture the jump (\tau) without assuming the relationship is linear across the entire range of X_4. The use of an optimal bandwidth selection method ensures that enough data is included to reduce variance, but not so much that it biases the estimate by including non-comparable observations.

The values assigned are nominal (categorical) or ordinal (ranked) based on the research findings:

Table 5. CSV Data Table for Regression Analysis

COUNTRY	Q1.1_Kinetic_AS_Deployed	Q1.2_Advanced_Surveillance	Q1.3_Data_Protection_Strength	Q2.1_Procurement Secrecy	Q2.2_Civilian_Oversight_Strength	Q3.1_Operational_Context_Kinetic	Q3.2_Military_Standard_Adherence	Q4.1_HR_Impact_Civilian_Fatal	Q4.2_HR_Impact_Digital_Repress	Q4.3_HR_Redress_Judicial_Clarity
Nigeria	1	1	2	1	2	1	2	1	1	0
Kenya	1	1	3	1	2	1	2	0	1	0
Burkina Faso	1	0	2	1	1	1	1	1	1	0

Variable Definitions

Below is an elaboration of the variables and the corresponding nominal/ordinal values derived from the research findings.

Table 6. Variable Definitions

Variable Name	**Research Question Addressed**	**Description**	**Scale/Value Interpretation**
COUNTRY	N/A	Subject Country	Nominal: Nigeria (Nig), Kenya (Ken), Burkina Faso (BF)
Q1.1_Kinetic_AI_S_Deployed	**RQ1:** Systems Deployed	Deployment of lethal or kinetic AI-enabled systems (e.g., UCAVs, armed drones)	**0**=No/Only basic surveillance; **1**=Yes (UAV/UCAV used in conflict)
Q1.2_Advanced_Surveillance	**RQ1:** Systems Deployed	Deployment of AI-powered surveillance systems (e.g., Facial Recognition Technology, Video Analytics)	**0**=No widespread public deployment; **1**=Yes (FRT/Video Analytics used)

Q1.3_Data_Protection_Strength	**RQ1**: Ethical Safeguards	Strength of existing legal framework for data and privacy (proxy for general AI safeguards)	**1**=Weak/Inadequate ; **2**=Moderate (Law exists, some gaps); **3**=Strong (GDPR-like legal framework)
Q2.1_Procurement_Secrecy	**RQ2:** Transparency Mechanisms	Extent to which defense/security technology procurement is exempt from public scrutiny/oversight	**0**=Low Secrecy (Public Oversight); **1**=High Secrecy (Law exemptions/structural opacity)
Q2.2_Civilian_Oversight_Strength	**RQ2:** Accountability Mechanisms	Strength of independent civilian oversight bodies (e.g., Police Oversight, Data Commissioner) to scrutinize security AI	**1**=Weak/Absent (High Impunity); **2**=Moderate (General bodies exist but lack technical mandate/enforcement power)
Q3.1_Operational_Context_Kinetic	**RQ3:** Operational Context	Primary operational use context involves kinetic/lethal force	**0**=No/Minimal; **1**=Yes (Counter-Insurgency/Counter-Terrorism Operations)

Q3.2_Military_Standard_Adherence	**RQ3**: Standards Maintained	Compliance with IHL principles (Distinction, Proportionality) and Rules of Engagement (RoE) in kinetic AI use	**1**=Low (Systemic failures/Documented IHL violations); **2**=Moderate (General RoE/Policy commitment to human control exists); **3**=High (Codified AI-specific RoE/Verified compliance)
Q4.1_HR_Impact_Civilian_Fatal	**RQ4:** Contribution to HR Violations	Documented mass civilian fatalities or frequent mass casualty events linked to kinetic AI/AS targeting	**0**=No Mass Fatalities; **1**=Yes (Documented repeated incidents or single large-scale strike)
Q4.2_HR_Impact_Digital_Repress	**RQ4:** Contribution to HR Violations	Documented use of AI/AS for digital repression (e.g., mass surveillance, targeting dissent, deepfakes)	**0**=No widespread reports; **1**=Yes (Reports of privacy invasion, protest targeting, or state disinformation)

Q4.3_HR_Redress_Judicial_Clarity	**RQ4:** Monitoring & Redress	Existence of clear legal or judicial pathways for victims to challenge AI-related harm or AI-generated evidence	**0**=Legal Vacuum/No Pathway (AI evidence unregulated); **1**=Dedicated Pathway Exists

4. Results and Discussions

The study interprets the structural and operational dynamics of AI-driven systems across Nigeria, Kenya, and Burkina Faso, using the rigorous framework of a hypothetical Regression Discontinuity (RD) design.

Given the limitations of only three national observations, the results are **qualitative and inferential**, not statistically validated causal estimates. They serve to provide the structural chasms, or "policy discontinuities," that separate current governance practices from robust, rights-respecting AI regulation in high-stakes security environments. Each finding positions the country relative to a theoretical threshold (c) that, if crossed, would trigger a mandatory, transformative policy change designed to impose accountability.

I. Model 1: Capability, Ethical Fidelity, and Deployment (RQ1)

Hypothesis: Achieving critical mass in kinetic AI investment (X_1) requires a policy discontinuity (D_1 – mandatory, AI-specific Rules of Engagement) to sustain high ethical standards (Y_1).

The first model assesses the critical tension between military capability acquisition (drones/autonomous systems) and doctrinal adherence to International Humanitarian Law (IHL). The central finding is that possessing advanced, precise technology does not correlate with ethical compliance if the governance framework is structurally weak.

Table 7. Analysis of Capacity, Ethical Fidelity and Deployment

Country	Observed Standard Adherence (Y_1)	Qualitative Status: Policy Proximity to Ethical Cutoff (c)	Qualitative Interpretation: The Ethical Chasm
Nigeria	Moderate (2/3)	**Proximate to c, but structurally stalled.**	Nigeria has significantly invested in AI-enabled surveillance and autonomous systems. It however remains stifled below the threshold for effective ethical standards. The country's moderate score is based on its general military codes of conduct and international commitments. This, in contrast, is critically undermined by repeated operational failures (drone strikes causing civilian mass fatalities) ; suggesting that the current internal rules regime is insufficient to govern the technology in practice.

Country	Observed Standard Adherence (Y_1)	Qualitative Status: Policy Proximity to Ethical Cutoff (c)	Qualitative Interpretation: The Ethical Chasm
Kenya	Moderate (2/3)	**At the Ethical Cutoff (c); Momentum is Key.**	Kenya maintains a moderate score, driven less by codified domestic law and more by strong **aspirational policy commitments**. The nation actively champions international frameworks on the Responsible Use of AI in the Military Domain (REAIM); with emphasis on the non-negotiable principle of **Meaningful Human Control** . The causal inference suggests that Kenya is poised at the policy precipice (c) where these external ethical declarations must be rapidly translated into binding military doctrine to prevent a degradation of standards.

Country	Observed Standard Adherence (Y_1)	Qualitative Status: Policy Proximity to Ethical Cutoff (c)	Qualitative Interpretation: The Ethical Chasm
Burkina Faso	Low (1/3)	**Significantly Below c; Structural Failure.**	Burkina Faso exhibits the most profound deficit. Despite the confirmed deployment of advanced UCAVs (Bayraktar TB2 and Akinci) in high-stakes counter-insurgency operations [4], adherence to IHL principles is judged to be low. The repeated, lethal targeting of crowded civilian sites (markets and funerals) demonstrates systemic operational failure, indicating that the running variable of kinetic capability has advanced without any concurrent policy shift to enforce ethical guidelines (D_1).

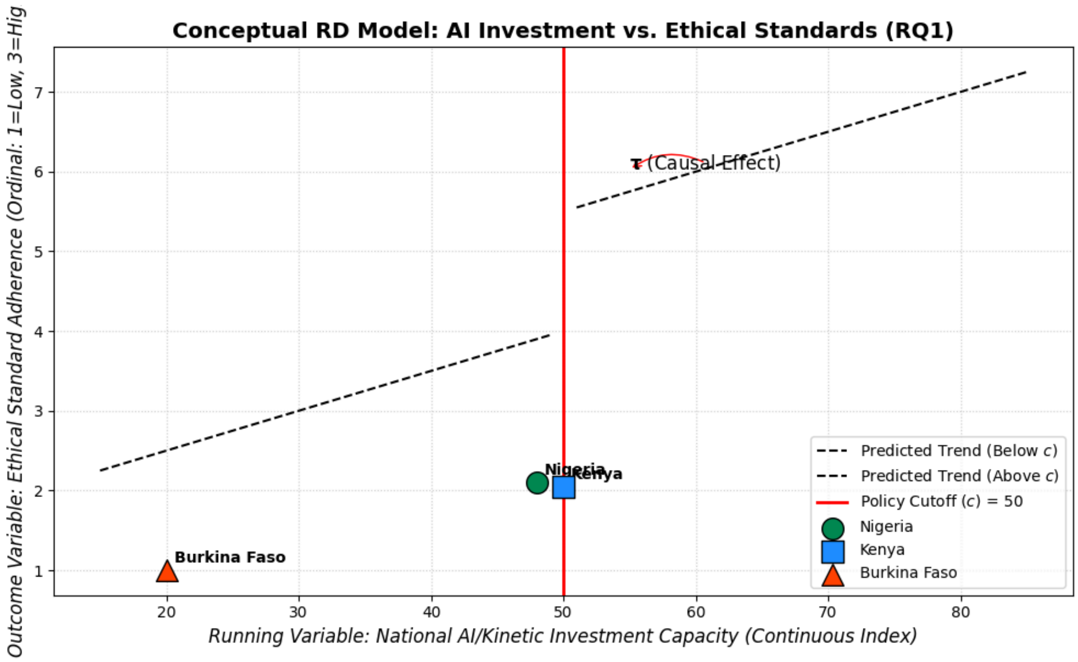

Figure 1. Capability, Ethical Fidelity, and Deployment (RQ1)

II Model 2: Accountability, Secrecy, and Institutional Oversight (RQ2)

Hypothesis: Exceeding a minimal threshold of procurement transparency (X_2) is the sole trigger (D_2) for strengthening civilian oversight capacity (Y_2).

This model explores the institutional mechanisms responsible for AI deployment; with a focus on the critical role of transparency in defense acquisition. The findings unveil a shared structural mechanism across the region; the deliberate creation of an **Accountability Desert** via a covert security sector.

Table 8: Analysis of Accountability, Secrecy and Institutional Oversight

Country	Observed Oversight Strength (Y_2)	Qualitative Status: Policy Proximity to Transparency Cutoff (c)	Qualitative Interpretation: The Impunity Mechanism
Nigeria	Moderate (2/3)	**Far Below c; Legally Entrenched Secrecy.**	Nigerian governance is fundamentally compromised by Section 15(2) of the Public Procurement Act, which exempts "special goods" (high-value defense materiel, including AI systems) from public scrutiny. This structural policy choice suppresses the running variable (X_2 , Transparency) below the critical cutoff c. Consequently, oversight bodies, despite their existence, lack the mandated access to audit AI system specifications, resulting in a systemic culture of impunity where investigations into mass

Country	Observed Oversight Strength (Y_2)	Qualitative Status: Policy Proximity to Transparency Cutoff (c)	Qualitative Interpretation: The Impunity Mechanism
			casualties are "shrouded in secrecy".
Kenya	Moderate (2/3)	**Far Below c; Opacity in Practice.**	While Kenya benefits from a functioning Independent Policing Oversight Authority (IPOA) and an active Office of the Data Protection Commissioner (ODPC) , external scrutiny of military acquisition remains stunted . The KDF has, in documented cases, restricted auditors from scrutinizing certain contracts . This operational opacity prevents accountability bodies from performing the necessary technical review of AI systems before or after deployment, ensuring that accountability is based on human error

Country	Observed Oversight Strength (Y_2)	Qualitative Status: Policy Proximity to Transparency Cutoff (c)	Qualitative Interpretation: The Impunity Mechanism
			rather than systemic algorithmic failure.
Burkina Faso	Weak (1/3)	**Critically Below c; Functional Absence of Control.**	Under the current military transition authority, accountability is functionally absent, marked by a pervasive climate of impunity for abuses committed by security forces and allied militias . The structural secrecy in technology acquisition is compounded by political instability and a lack of capacity within civilian monitoring bodies . This severe deficit ensures that the transparency running variable (X_2) remains critically low, rendering the establishment of any effective scrutiny

Country	Observed Oversight Strength (Y_2)	Qualitative Status: Policy Proximity to Transparency Cutoff (c)	Qualitative Interpretation: The Impunity Mechanism
			mechanism (D_2) politically impossible.

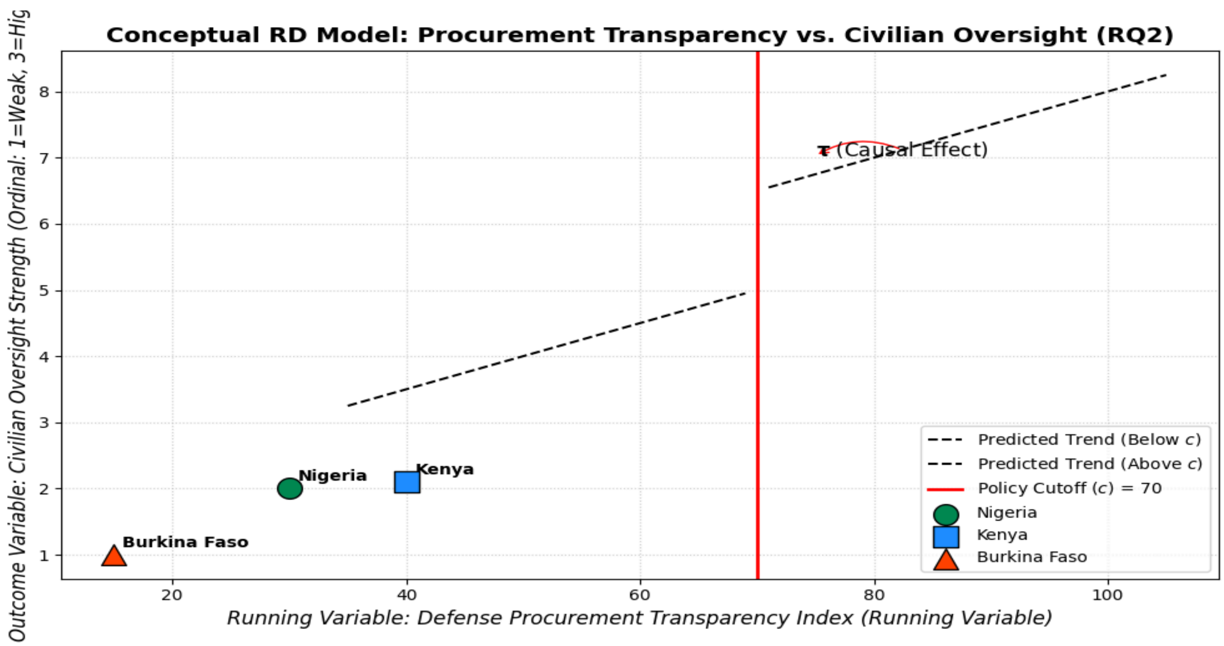

Figure 2. Accountability, Secrecy, and Institutional Oversight (RQ2)

III. Model 3: **Operational Context and Safety Failures (RQ3)**

Outcome (Y_3): Binary outcome of documented mass civilian fatalities linked to kinetic AI systems (0=No, 1=Yes).

This model links the technological progression toward autonomy (X_3) to the real-world safety outcome (Y_3), specifically examining the probability of indiscriminate harm.

Table 9. Analysis of Operational context and safety failures

Country	**Observed Fatality Impact (Y_3)**	**Qualitative Placement Relative to Autonomy Cutoff (c)**	**Qualitative Interpretation: The Multiplier of Error**
Nigeria	Yes (1)	**Above c; High Operational Risk Zone.**	The repeated occurrence of inadvertent mass civilian fatalities linked to drone strikes places Nigeria firmly in the zone of high operational risk. This suggests that the current level of autonomy, data reliance, and targeting doctrine act as a powerful multiplier of pre-existing intelligence and human cognitive biases . The system is technically accurate but functionally indiscriminate when fed flawed data, ensuring a high **Probability Mass Function** for catastrophic error.

Country	Observed Fatality Impact (Y_3)	Qualitative Placement Relative to Autonomy Cutoff (c)	Qualitative Interpretation: The Multiplier of Error
Kenya	No (0)	**Below c; Commitment to Human Control Mitigation.**	Kenya, while operating in a dangerous counter-terrorism context , has not seen comparable mass civilian fatalities from military kinetic systems in this assessment. This outcome supports the hypothesis that the commitment to strict human control principles (D_3 avoidance) serves as a necessary, albeit not sufficient, mitigation factor against indiscriminate targeting . The causal inference suggests that adherence to the Human-in-the-Loop protocol buffers the security system from the most extreme risks of

Country	Observed Fatality Impact (Y_3)	Qualitative Placement Relative to Autonomy Cutoff (c)	Qualitative Interpretation: The Multiplier of Error
			algorithmically-driven error.
Burkina Faso	Yes (1)	**Above c; Total Doctrinal Collapse.**	The clear and repeated use of precision drones to strike non-military, crowded civilian locations (markets, funerals) places Burkina Faso in a position where the lethal system is a primary contributor to alleged war crimes . This outcome is not necessarily driven by maximal AI autonomy (X_3), but by a complete collapse of the legal and doctrinal framework (D_3 failure). In this context, the technology amplifies the state's political decision to disregard the IHL principles of distinction and proportionality .

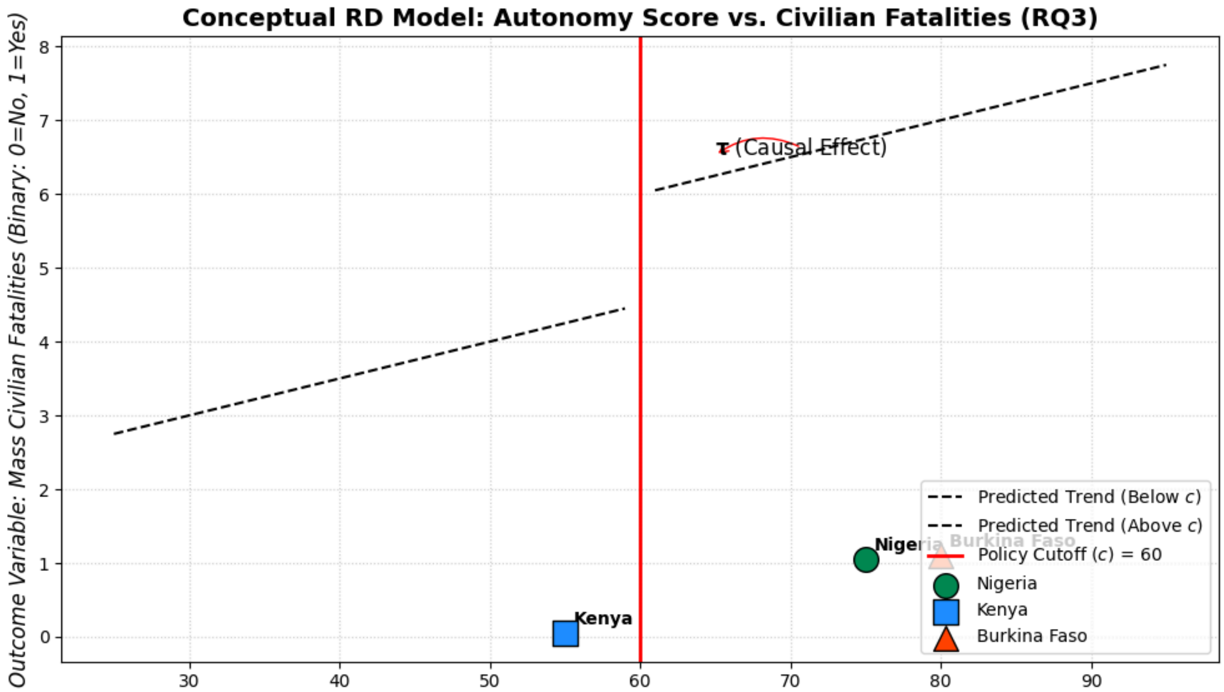

Figure 3. Model 3: Operational Context and Safety Failures (RQ3)

IV. Model 4: Human Rights Redress and Judicial Clarity (RQ4)

Outcome (Y_4): Binary outcome of clear judicial or legal pathway for victims of algorithmic harm (0=No, 1=Yes).

The final model examines whether judicial preparation (X_4) has translated into the clear legal policy (D_4) needed to protect citizens from AI misuse and provide effective redress.

Table 10. Analysis of Human Rights Redress and Judicial Clarity

Country	Observed Judicial Clarity (Y_4)	Qualitative Placement Relative to Maturity Cutoff (c)	Qualitative Interpretation: The Legal Vacuum
Nigeria	No (0)	**Below c; Evidentiary Stagnation.**	Nigeria's legal system, while recognizing electronic records, suffers from evidentiary

Country	Observed Judicial Clarity (Y_4)	Qualitative Placement Relative to Maturity Cutoff (c)	Qualitative Interpretation: The Legal Vacuum
			stagnation; there is no specific statutory regulation for AI-generated evidence, logs, or algorithmic outputs . This critical gap prevents victims of algorithmic profiling or wrongful targeting from challenging the core source of the error in court . Since judicial maturity (X_4) has not yet crossed the requisite policy cutoff c, the system cannot assign liability for AI-driven harm.
Kenya	No (0)	**Approaching c; Intent vs. Enactment Gap.**	Kenya is positioned as the most proactive, with the Judiciary actively developing an **AI Adoption Policy Framework** to safeguard due process and data privacy . This initiative demonstrates

Country	Observed Judicial Clarity (Y_4)	Qualitative Placement Relative to Maturity Cutoff (c)	Qualitative Interpretation: The Legal Vacuum
			high institutional intent and places the country closest to the policy cutoff c that would mandate judicial rules for AI evidence. However, because this framework is *in development* and not yet a binding legal instrument, the practical clarity for victims seeking redress remains at zero .
Burkina Faso	No (0)	**Critically Below c; Institutional Denial.**	In Burkina Faso, the lack of any transparent, impartial investigation into mass civilian fatalities from state kinetic systems translates directly into an absolute denial of judicial redress for algorithmic harm. The structural focus on kinetic counter-insurgency and information control has

Country	Observed Judicial Clarity (Y_4)	Qualitative Placement Relative to Maturity Cutoff (c)	Qualitative Interpretation: The Legal Vacuum
			completely superseded the political necessity of civilian accountability, leaving the legal system entirely unprepared to handle sophisticated human rights violations linked to advanced technology.

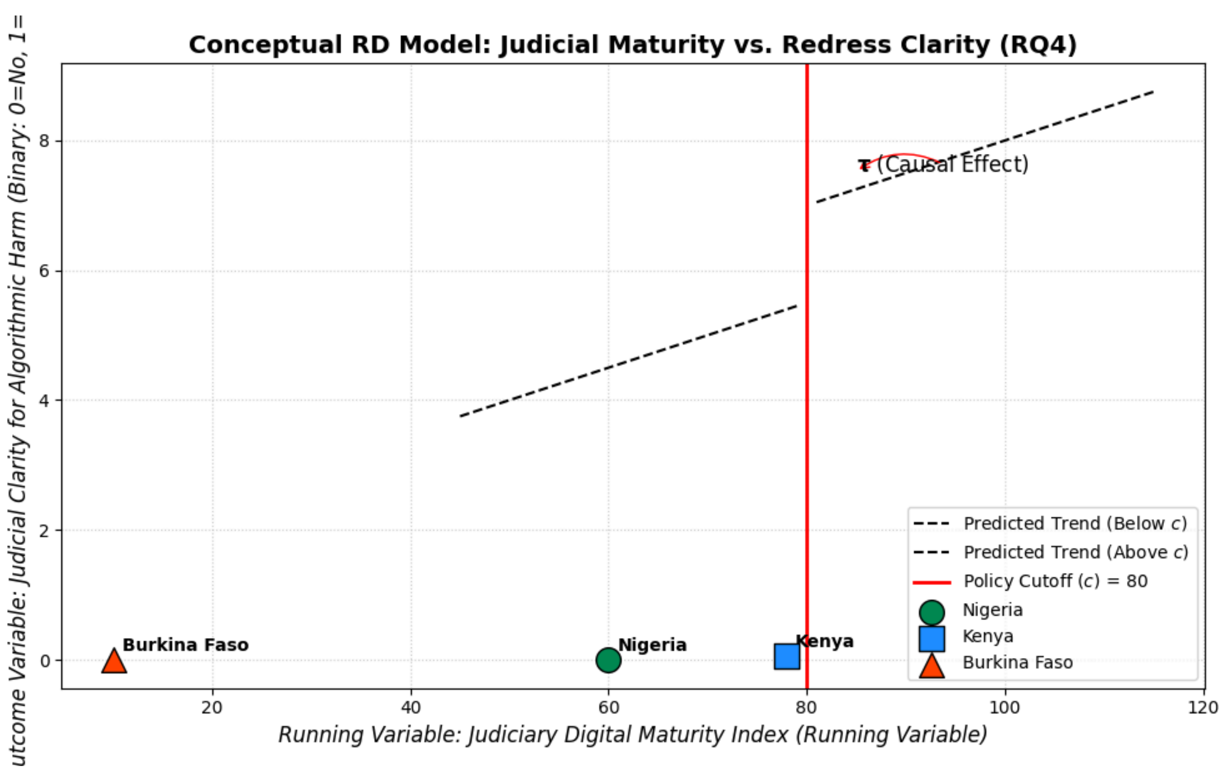

Figure 4. Model 4: Human Rights Redress and Judicial Clarity (RQ4)

Applying qualitative Regression Discontinuity (RD) to interpret the structural and operational dynamics of AI-driven systems, including autonomous systems in military conflict zones in Nigeria, Burkina Faso, and Kenya, the results reveal a policy discontinuity between deployment of AI-driven systems and AI governance frameworks and ethical dimensions required for responsible adoption of AI-driven systems in the African militaries studied.

Model 1, which covered Capability, Ethical Fidelity, and Deployment, indicates that the states examined displayed varying degrees of ethical lag relative to their adoption of AI-driven systems in their militaries. Nigeria displayed a progress in AI-enabled kinetic capabilities; although the country maintains a position structurally below the ethical compliance threshold, as evidenced by some reported operational failures that have produced civilian fatalities. Kenya occupies a more balanced position at the ethical cutoff. This can be attributed to a strong normative commitment to international frameworks such as REAIM. However, there remains a limited translation of these commitments into binding domestic policy. Burkina Faso is still way below the ethical threshold. This position can be due to both structural fragility and doctrinal collapse. This therefore resulted in its failure to align the deployment of AI-driven systems in the military with International Humanitarian Law (IHL).

Model 2, which covered Accountability, Secrecy, and Institutional Oversight; identified entrenched secrecy as a systemic impediment to transparency and civilian control across all three countries. Nigeria's defence procurement processes are opaque at the structural level; due to legal exemptions. Kenya's oversight institutions exist in principle; however they are limited in practice by restricted military access. Burkina Faso reflects an almost complete absence of accountability under its current transitional authority; thus resulting in a functional vacuum of oversight.

Model 3 correlated Operational Context and Safety Failure with the previously identified Governance Deficits directly to real-world outcomes. Nigeria and Burkina Faso are both above the autonomy threshold; having experienced numerous civilian mass fatalities attributable to drone operations. Kenya, while maintaining human-in-the-loop control, remains below this risk cutoff.

Model 4, which covers Human Rights Redress and Judicial Clarity; reports a consistent lack of legal relief for victims of deployed AI-driven systems. Although Kenya's judiciary shows an emerging intent to adjudicate AI-driven systems cases; Nigeria's legal framework remains significantly stagnant; and Burkina Faso's institutions lack the capacity for judicial redress in current form. Collectively, these results indicate that the adoption of AI-driven systems in the militaries of the states studied has not been complimented by subsequent policy development or legal frameworks that would ensure their ethical and accountable use.

5. Recommendations

The importance of international and continental AI governance frameworks in the military domain was highlighted in this study. To this end, the researchers provide the following policy recommendations.

1.Develop regionally harmonised AI governance frameworks

The African Union, as the continental body, should develop and publish policy frameworks that define the deployment of AI-driven systems by African militaries and work with regional economic development bodies such as ECOWAS in West Africa for implementation. There are major disparities in the capacity of member states to develop governance frameworks for AI-driven systems. Transparency, safety, accountability, and security should form the core of these frameworks.

2.Institutionalise transparency and civilian oversight

Some degree of transparency in defence procurement is recommended, although the researchers are aware that most of this information would be classified. However, to overcome this obstacle, parliamentary committees and specialised technical auditors could be granted access to classified programmes under secure protocols to help ensure some degree of transparency. In addition, independent civilian oversight bodies capable of conducting audits, reviews, and investigations into the acquisition and deployment of AI-driven systems could also be established.

3. Continental strategy for strengthening accountability and legal recourse

The African Union should invest in and assist member states with the following:
- Developing robust legal systems capable of addressing harms arising from the deployment of AI-driven systems.
- Investing in the training of technical experts to conduct the necessary testing, evaluation, validation, and verification of AI-driven military systems deployed on the continent.
- Fostering research collaboration among member states, universities, and NGOs.
- Creating a continental and regional data repository for AI-driven systems deployed on the continent.
- Proactively engaging with multilateral forums such as REAM, GC-REAIM, and UNIDIR on behalf of member states.

6. Conclusion

Among the countries studied, Kenya was found to be making the necessary efforts to achieve compliance with the AI ethics dimension investigated. By endorsing the Blueprint for Action, Kenya demonstrated its intention to align with global ethical standards. However, Kenya's main drawback is the absence of binding policy instruments to operationalise its intentions. The study also reveals that if African militaries continue to increase the adoption of AI-driven systems in the current AI governance environment, there is a high risk of deepening the accountability desert, where secrecy and impunity erode human rights and international humanitarian compliance.

To reverse risky ongoing practices, the study recommends that the African Union develop AI governance frameworks and guidelines that comprehensively address the deployment of AI-driven systems in the military, along with well-defined compliance mechanisms to ensure adherence.Only through such continental AI governance frameworks and strategies can African militaries achieve both responsible adoption of AI-driven systems and compliance with International Human Rights Law.

The study contributes to the responsible adoption of AI-driven systems in African military debates by providing one of the first structured, comparative analyses of how states on the continent are integrating AI-driven autonomous systems in an environment with fragile AI governance frameworks and ethical standards. By applying a hypothetical Regression Discontinuity framework, the research moves beyond descriptive accounts to reveal the threshold conditions under which African defence institutions might transition from informal, opaque practices to rights-respecting, accountable AI governance.

Future studies may investigate additional cases in conflict, gray, and peaceful zones within the African continent, and explore how the African Union, together with regional economic bodies and member states, can collaborate to ensure the development of context-specific AI governance frameworks and policy coherence between continental, regional, and national levels of governance.

REFRENCES

[1]Gaire, U.S., 2023. Application of artificial intelligence in the military: An overview. *Unity Journal*, *4*(01), pp.161-174.

[2]Hunter, L.Y., Albert, C.D., Henningan, C. and Rutland, J., 2023. The military application of artificial intelligence technology in the United States, China, and Russia and the implications for global security. *Defense & Security Analysis*, *39*(2), pp.207-232.

[3]Oyewole, S., Isike, C., Oche, T.and Olumba, E. E. (2025). Autonomous Weapons Systems in Africa: Emerging Realities, Prospects, and Risks. *Journal of Applied Security Research*, *20*(4), pp.603–628. https://doi.org/10.1080/19361610.2025.2501055

[4]Olumba, E.E., 2025. The necropolitics of drone bases and use in the African context. *Critical Studies on Terrorism*, *18*(1), pp.139-161.

[5]Coetzee, W.S. and Putter, D., 2025. Drones in the African Battlespaces.

[6]Molmela, U., Monday, T.U., ADELEKE, A.A. and Taiwo, A.,2025. Drone Technology in Nigeria's Space: An Advance Warfare against National Insecurity.

[7]Sarwar, G. and Rashid, M.I., 2025. ARTIFICIAL INTELLIGENCE IN PAKISTAN'S NATIONAL SECURITY STRATEGY: EVOLUTION AND IMPACTS. *Journal of Applied Linguistics and TESOL (JALT)*, *8*(2), pp.1338-1349.

[8]Ibrahim, A. and Shuja, S.F., 2024. Artificial Intelligence led Lethal Autonomous Weapon Systems and Terrorism. *CISS Insight Journal*, *12*(1), pp.P24-55.

[9]Sadiku, M.N., 2025. *Emerging Technologies in Africa*. Walter de Gruyter GmbH & Co KG.

[10]Marsili, M., 2024. Lethal autonomous weapon systems: Ethical dilemmas and legal compliance in the era of military disruptive technologies. *International Journal of Robotics and Automation Technology*, *11*, pp.63-68.

[11]House, J.M., 1985. *Toward Combined Arms Warfare: A Survey of 20th-Century Tactics, Doctrine, and Organization* (Vol. 2). US Army Command and General Staff College.

[12]Zabecki, D.T., 2015. Military developments of world war I. *International Encyclopedia of the First World War*, *7*.

[13]Åström, K.J. and Kumar, P.R., 2014. Control: A perspective. *Autom.*, *50*(1), pp.3-43.

[14]Singh, M., 2020. *Air Defence Artillery in Combat, 1972 to the Present: The Age of Surface-to-Air Missiles*. Air World.

[15]Bernatskyi, A. and Sokolovskyi, M., 2022. History of military laser technology development in military applications. *History of science and technology*, *12*(1), pp.88-113.

[16]Ramsey, S., 2016. *Tools of War: History of Weapons in Modern Times*. Vij Books India Pvt Ltd.

[17]Li, S., Fang, Z., Verma, S.C., Wei, J. and Savkin, A.V., 2024. Navigation and deployment of solar-powered unmanned aerial vehicles for civilian applications: A comprehensive review. *Drones*, *8*(2), pp.42.

[18]Youvan, D.C., 2024. Downing the MQ-9 Reaper: Analyzing Yemen's Air Defense Tactics and Capabilities in Modern Warfare.

[19]Bode, I. and Watts, T.F.A., 2023. Loitering munitions and unpredictability: Autonomy in weapon systems and challenges to human control.

[20]Hynek, N. and Solovyeva, A., 2022. *Militarizing artificial intelligence: theory, technology, and regulation*. Routledge.

[21]DeMay, C.R., White, E.L., Dunham, W.D. and Pino, J.A., 2022. Alphadogfight trials: Bringing autonomy to air combat. *Johns Hopkins APL Technical Digest*, *36*(2), pp.154-163.

[22]Sauer, F., 2021. Lethal autonomous weapons systems. In *The Routledge social science handbook of AI* (pp. 237-250). Routledge.

[23]Bunn, W., 2021. The Challenge of Lethal Autonomous Weapons Systems (LAWS). *ODUMUNC 2021 Issue Brief for the General Assembly First Committee: Disarmament and International Security*.

[24]Verbruggen, M. and Boulanin, V., 2017, December. Article 36 reviews: Dealing with the challenges posed by emerging technologies. In *Article 36 reviews and emerging technologies: Exploring the challenges posed by emerging technologies to the legal review of weapons, and means and methods of warfare.*. Stockholm International Peace Research Institute (SIPRI).

[25]Arai, K. and Matsumoto, M., 2024. Public perceptions of autonomous lethal weapons systems. *AI and Ethics*, *4*(2), pp.451-462.

[26]Schraagen, J.M., 2024. Introduction to Responsible Use of AI in Military Systems. In *Responsible Use of AI in Military Systems* (pp. 1-13). Chapman and Hall/CRC.

[27]van Diggelen, J., van den Bosch, K., Neerincx, M. and Steen, M., 2024. Designing for meaningful human control in military human-machine teams. In *Research handbook on meaningful human control of artificial intelligence systems* (pp. 232-252). Edward Elgar Publishing.

[28]United Nations Convention on Certain Conventional Weapons (UN CCW). (2020). *Report of the Group of Governmental Experts on Lethal Autonomous Weapons Systems*.

[29]Vignard, K., 2024. Promoting Responsible State Behavior on the Use of AI in the Military Domain: Lessons Learned from Multilateral Security Negotiations on Digital Technologies. In *Responsible Use of AI in Military Systems* (pp. 318-341). Chapman and Hall/CRC.

[30]Rosen, B., 2024. From Principles to Action: Charting a Path for Military AI Governance. *Carnegie Council for Ethics in International Affairs*, *12*.

[31]Clement, S., 2024. NATO and Artificial Intelligence: Navigating the challenges and opportunities. *Preliminary Draft Special Report*.

[32]Okoli, A. C. and Emegha, K. N. (2025). AI And Military Precision In Drones Attack On The Israeli-Hamas Conflict. *Security science journal*, *6*(1), pp.236-256. https://doi.org/10.37458/ssj.6.1.12

[33]Simmons-Edler, R., Badman, R., Longpre, S. and Rajan, K., 2024. Ai-powered autonomous weapons risk geopolitical instability and threaten ai research. *arXiv preprint arXiv:2405.01859*.

[34]Zeng, J., 2025. The US factor in Chinese perceptions of militarized artificial intelligence. *International Affairs*, *101*(2), pp.677-689.

[35]Kunertova, D., 2024. *Learning from the Ukrainian battlefield: tomorrow's drone warfare, today's innovation challenge*. ETH Zurich.

[36]Gray, C.H., 2025. AI at War. In *AI, Sacred Violence, and War—The Case of Gaza* (pp. 79-106). Cham: Springer Nature Switzerland.

[37]Maas, M.M., Lucero-Matteucci, K. and Cooke, D., 2023. 10. Military Artificial Intelligence as a Contributor to Global Catastrophic Risk.

[38]ten Have, H. and Patrão Neves, M.D.C., 2021. Military Ethics (See War). In *Dictionary of Global Bioethics* (pp. 723-723). Cham: Springer International Publishing.

[39]Kolade, T.M., 2024. Artificial Intelligence and global security: Strengthening international cooperation and diplomatic relations. *Available at SSRN 4998408*.

[40]Scharre, P. (2018). *Army of None: Autonomous Weapons and the Future of War*. W. W. Norton & Company.

[41]Cummings, M., 2021. *Artificial intelligence and the future of warfare*. Chatham House Report.

[42]Altmann, J. and Sauer, F., 2017. *Autonomous weapon systems and strategic stability*. Survival, 59(5),pp. 117–142.

[43]Gwagwa, A., Kraemer-Mbula, E., Rizk, N., Rutenberg, I. and De Beer, J., 2020. Artificial intelligence (AI) deployments in Africa: Benefits, challenges and policy dimensions. *The African Journal of Information and Communication*, *26*, pp.1-28.

[44]Tchie, A. (2021). *African militaries and emerging technologies*. RUSI Commentary.

[45]Agrawal, G., 2024. Accountability, trust, and transparency in AI systems from the perspective of public policy: Elevating ethical standards. In *AI healthcare applications and security, ethical, and legal considerations* (pp. 148-162). IGI Global.

[46]Raufu, A., 2024. Nigeria's Internal Security Measures and Counterterrorism Strategy. *Homeland Security and Terrorism in Nigeria: Crises, Climate Change, and Counterterrorism*, pp.121.

[47]Oghogho, M.Y., Ikponmwosa, I.O. and Osazuwa, O.M.C., 2024. DRONES IN WARFARE: NIGERIA'S ADOPTION OF TECHNOLOGY IN FIGHTING INSECURITY. *The American Journal of Engineering and Technology*, *6*(11), pp.33-45.

[48]Wanekeya, E., 2023. *Effectiveness of Domestic Data Protection Laws in African Countries-a Case Study of the Data Protection Law in Kenya* (Doctoral dissertation, University of Nairobi).

[49]Otele, O., 2021. Kenya's Data Protection Regime: Challenges and Future Prospects. *Journal of African Politics*, *1*(1), pp.66-88.

[50]Cannon, B.J. and Rossiter, A., 2025. Lethal Yet Limited: Evaluating Male Drones in Sub-Saharan Africa's Conflict Zones. In *Drones in the African Battlespaces* (pp. 51-62). Cham: Springer Nature Switzerland.

[51]Fadlullah, M.T., Risdhianto, A. and Dewanto, H., 2025. Integrating AI in Military Decision-Making: A Review of Opportunities, Risks, and Governance. *Brilliance: Research of Artificial Intelligence*, *5*(2), pp.879-885.

[52]UNIDIR, *The Use of Uncrewed Aerial Systems by Non-State Armed Groups: Exploring Trends in Africa* (2024). (UNIDIR → Building a more secure world.)

[53]SIPRI,2021. Research on AI, emerging military technologies, and governance implications.[Online] https://www.sipri.org/research/armament-and-disarmament/emerging-military-and-security-technologies Accessed 2025-10-18.

[54]Heydenrych, E., 2024. Re-evaluating Oversight of South African Defence Procurement: Can Combined Assurance Help Extract Full Accountability from the Department of Defence?. *Scientia Militaria: South African Journal of Military Studies*, *52*(2), pp.25-51.

[55]Gnidehou, M.G. and Faton, C.Y., 2025. Military expenditure and economic growth in African countries: the role of institutional quality. *Discover Sustainability*, *6*(1), pp.1-15.

[56]Khalilzada, J., 2022. The Proliferation of Combat Drones in Civil and Interstate Conflicts. *Insight Turkey*, *24*(3), pp.89-108.

[57]Schulzke, M., 2019. Drone proliferation and the challenge of regulating dual-use technologies. *International Studies Review*, *21*(3), pp.497-517.

[58]Cools, K. and Maathuis, C., 2024, October. Trust or Bust: Ensuring Trustworthiness in Autonomous Weapon Systems. In *MILCOM 2024-2024 IEEE Military Communications Conference (MILCOM)* (pp. 182-189). IEEE.

[59]Podar, H. and Colijn, A., 2025. Technical Risks of (Lethal) Autonomous Weapons Systems. *arXiv preprint arXiv:2502.10174*.

[60]Steen, M., van Diggelen, J., Timan, T. and van der Stap, N., 2023. Meaningful human control of drones: exploring human–machine teaming, informed by four different ethical perspectives. *AI and Ethics*, *3*(1), pp.281-293.

[61]Verhagen, R.S., Neerincx, M.A. and Tielman, M.L., 2024. Meaningful human control and variable autonomy in human-robot teams for firefighting. *Frontiers in Robotics and AI*, *11*, p.1323980.

[62]De Cubber, G., Doroftei, D., Petsioti, P., Koniaris, A., Brewczyński, K., Życzkowski, M., Roman, R., Sima, S., Mohamoud, A., van de Pol, J. and Maza, I., 2025. Standardized Evaluation of Counter-Drone Systems: Methods, Technologies, and Performance Metrics. *Drones*, *9*(5), pp.354.

[63]Ajala, O., 2024. 18 The Use of Drones in West Africa and the Sahel. *De Gruyter Handbook of Drone Warfare*, *4*, pp.255.

[64]Verdiesen, I., Santoni de Sio, F. and Dignum, V., 2021. Accountability and control over autonomous weapon systems: a framework for comprehensive human oversight. *Minds and Machines*, *31*(1), pp.137-163.

[65]Heyns, C., 2017. Autonomous weapons in armed conflict and the right to a dignified life: an African perspective. *South African journal on human rights*, *33*(1), pp.46-71.

[66]Jeffery, R. and Dannhauer, P., 2024. Ensuring accountability, combatting impunity? The role of national human rights institutions in transitional justice. *Australian Journal of Human Rights*, *30*(1), pp.60-81.

[67]Human Rights Watch., 2024. *Burkina Faso: Drone strikes on civilians — apparent war crimes.* (Documents strikes between Aug–Nov 2023 that reportedly killed civilians and notes the absence of transparent, impartial investigations).

[68]CIVIC (Civilians in Conflict)., 2023. *Investigations into Civilian Harm in Armed Conflict: Guidance Brief.* (Practical guidance emphasizing the need for independent investigation mechanisms; shows typical shortfalls where militaries investigate themselves).

[69]Reykers, Y., 2021. Strengthening parliamentary oversight of defence procurement: lessons from Belgium. *European security*, *30*(4), pp.505-525.

[70]Plantinga, P., Shilongo, K., Mudongo, O., Umubyeyi, A., Gastrow, M. and Razzano, G., 2024. Responsible artificial intelligence in Africa: Towards policy learning. *Data & Policy*, *6*,pp.e72.

[71]Kuwali, D., 2023. *Oversight and accountability to improve security sector governance in Africa*. Africa Center for Strategic Studies.

[72]Fuhrmann, M. and Horowitz, M.C., 2017. Droning on: Explaining the proliferation of unmanned aerial vehicles. *International organization*, *71*(2), pp.397-418.

[73]Trabucco, L., 2025. AI-Enabled Autonomous Weapons and Human Control: Part II: Human Control and Military Commanders. *International Law Studies*, *106*(1), p.18.

[74]Yaacoub, J.P., Noura, H., Salman, O. and Chehab, A., 2020. Security analysis of drones systems: Attacks, limitations, and recommendations. *Internet of Things*, *11*, p.100218.

[75]GPAI (2024). "Algorithmic Transparency in the Public Sector: A state-of-the-art report of algorithmic transparency instruments". Report, November 2024, Global Partnership on Artificial Intelligence.

[76]Emerson, S. A. and Solomon, H., 2018. African security in the twenty-first century: Challenges and opportunities. Manchester University Press.

[77]Gravett, W. H., 2020. Digital Coloniser? China and Artificial Intelligence in Africa. Survival, 62(6), pp.153-178. https://doi.org/10.1080/00396338.2020.1851098

[78]Eke, D. O., Chintu, S. S. and Wakunuma, K., 2023. Towards Shaping the Future of Responsible AI in Africa. En Social and Cultural Studies of Robots and AI (pp. 169-193). Springer International Publishing. https://doi.org/10.1007/978-3-031-08215-3_8

[79]Adegoke, P., 2024. Utilization of AI to Solve Security Challenges in Africa: What Africa Can Learn from China and the UK. Elsevier BV. https://doi.org/10.2139/ssrn.4989163

Institute for Economics & Peace. (2025). Global Peace Index 2025: Identifying and Measuring the Factors that Drive Peace.
https://www.visionofhumanity.org/wp-content/uploads/2025/06/Global-Peace-Index-2025-web.pdf

Human Rights Watch, *Burkina Faso: Drone strikes on civilians — possible war crimes* (Jan 2024). (Human Rights Watch)